

The Crystal Realm of LUMINARA

Queen Auralia rules the **Crystal Realm of LUMINARA**, residing in the **Sunspire Palace** along with Prince Solen. The **Scholars of Light**, powerful technomancies, live in **Photon Cove**, where they operate the Realm's central data archives and communication towers.

The leader of the Scholars, **Archon Lytros**, believes technology should only be used to preserve knowledge and protect peace. The **Stonefolk**, who are miners and craftsmen, live deep within the **Granite Borough**, where they construct structures for the kingdom.

Beyond the eastern sea lies the **Corsair Isles**, a dangerous region controlled by sea raiders. Though most corsairs attack merchant ships, some launch raids on Granite Borough to capture Stonefolk for forced labor. The corsairs maintain their own underground communication network and never directly coordinate with the Crystal Realm.

The Scholars of Light secretly assist in rescuing captured Stonefolk and manage the **Frostfield Ranch**, where cold-climate creatures are bred for food and transport. Prince Solen enjoys racing crystal sleds with the Stonefolk at **Aurora Ridge**.

One day, Queen Auralia commands Archon Lytros to establish a **reliable communication network** across all regions of LUMINARA. However, Archon Lytros refuses to activate the ancient magic-powered network, stating that knowledge must flow through **modern technological systems instead of magic**.

Thus, **you are appointed as the Chief Network Architect** of the Crystal Realm.

Archon Lytros provides you with the list of locations and distances. Your task is to **design and deploy a complete network infrastructure** so that all regions can communicate efficiently.

Locations and Population

Location	Population
Sunspire Palace	45
Photon Cove	80
Frostfield Ranch	350
Corsair Isles	1200
Granite Borough	5200
Aurora Ridge	150

Distance Matrix (in miles)

From / To	Sunspire Palace	Photon Cove	Frostfield Ranch	Corsair Isles	Granite Borough	Aurora Ridge
Sunspire Palace (45)	0	280	510	160	720	240
Photon Cove (80)	280	0	190	260	60	530
Frostfield Ranch (350)	510	190	0	470	340	70
Corsair Isles (1200)	160	260	470	0	30	890
Granite Borough (5200)	720	60	340	30	0	110
Aurora Ridge (150)	240	530	70	890	110	0

Network Design Rules & Constraints

While designing the network infrastructure, the following restrictions must be strictly followed:

1. **Each location must be treated as a separate network connected by routers.**
2. Select the student ID of the last member of your group. Now take the last 4 digits of that ID and divide them into two parts containing two digits each. These two parts are the first two octets of the network address of the location. The third and fourth octets are 0, and the prefix mask is 16. Suppose the ID is 20201002. Then the last four digits are 1002. Now dividing it into two parts gives 10 and 02. So, the Network address is 10.2.0.0/16.

You must:

- o Perform VLSM subnetting
- o Assign one unique subnetwork address for each location

3. **Assign valid IP addresses to:**

- All router interfaces
- At least **two end devices per location**

4. **Photon Cove must contain:**

- A **Web Server**
- A **DNS Server**
- An **Email Server**

The web server must display the homepage message: “**Welcome to the Crystal Realm of LUMINARA**” when accessed from **any location**, using the domain: www.luminara.net

Email service is available only among the users of **Sunspire Palace, Photon Cove, and Granite Borough**

Shortest Path Rule:

- Establish connections among all networks using the **shortest possible routes based on the given distances** considering **Photon Cove** as root.

5. **Routing Requirements:**

- There must be **one primary route from Sunspire Palace to Granite Borough and one floating static route to same destination via Aurora Ridge**.
- **No default routes are allowed** anywhere in the system.
- **Static routing must be configured only between:**
 - Sunspire Palace ↔ Photon Cove (using recursive static route)
 - Photon Cove ↔ Granite Borough (using directly attached exit interface)
 - Sunspire Palace ↔ Aurora Ridge (using recursive static route)
- All **other routes must be configured using dynamic routing** (RIP v2).

6. **DHCP Requirement:**

- All IP addresses for **Frostfield Ranch** and **Aurora Ridge** must be assigned using **DHCP**.
- The **DHCP Server is located in Photon Cove LAN**.

7. Final Verification:

- After implementation, you must ensure:
 - Successful **ping from any device in one location to any device in another location**
 - Successful **web access to www.luminara.net from all networks**
 - Successful email service among two users of different LANs.

Software Requirement

The entire network must be implemented in Cisco Packet Tracer with:

- Routers
- Switches
- End devices
- Servers
- Proper cabling
- Full IP configuration
- Routing protocols fully functional

Final Deliverables

You will have to submit the following:

- Work Distribution among the group members [Who did which part]
- The pkt/pka file
- Picture of the Network topology diagram with proper labels [You have to show the network addresses using notes for each network]
- A PDF containing
 - VLSM tree
 - IP address table
 - The configuration commands of all the routers you have implemented.